

which is why there is no n in $O(1)$. You might also see constant time written as $O(c)$.

$O(\log n)$, Logarithmic

Logarithms are the inverse of exponentiation: the exponent 2^4 , or $2 \times 2 \times 2 \times 2$, equals 16, whereas the logarithm $\log_2(16)$ (pronounced “log base 2 of 16”) equals 4. In programming, we often assume base 2 to be the logarithm base, which is why we write $O(\log n)$ instead of $O(\log_2 n)$.

Searching for a book on an alphabetized bookshelf is a logarithmic time operation. To find one book, you can check the book in the middle of the shelf. If that is the book you’re searching for, you’re done. Otherwise, you can determine whether the book you’re searching for comes before or after this middle book. By doing so, you’ve effectively reduced the range of books you need to search in half. You can repeat this process again, checking the middle book in the half that you expect to find it. We call this the *binary search algorithm*, and there’s an example of it in “Big O Analysis Examples” later in this chapter.

The number of times you can split a set of n books in half is $\log_2 n$. On a shelf of 16 books, it will take at most four steps to find the right one. Because each step reduces the number of books you need to search by one half, a bookshelf with double the number of books takes just one more step to search. If there were 4.2 billion books on the alphabetized bookshelf, it would still only take 32 steps to find a particular book.

Log n algorithms usually involve a *divide and conquer* step, which selects half of the n input to work on and then another half from that half, and so on. Log n operations scale nicely: the workload n can double in size, but the runtime increases by only one step.

$O(n)$, Linear Time

Reading all the books on a bookshelf is a linear time operation. If the books are roughly the same length and you double the number of books on the shelf, it will take roughly double the amount of time to read all the books. The runtime increases *in proportion* to the number of books n .